

HLD Learning Guide

A Complete High-Level System Design Program

From zero HLD knowledge to interview-ready
Built for Software Engineers preparing for Tier-1 & Tier-2 product companies
India • UAE / Dubai • Saudi Arabia

Prepared for: Wasim
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How to Use This Guide

This is the master index for your System Design (HLD) preparation program. It does not contain full chapter content — think of it as the syllabus and study planner. Full, detailed content is delivered phase by phase as separate chapter PDFs, generated only when you approve that phase.

The program has 4 phases, moving from absolute basics to interview-day performance:

- **Phase 1 — Fundamentals:** the building blocks. Assumes zero prior HLD knowledge.
- **Phase 2 — Intermediate:** scalability, consistency, sharding, messaging, microservices basics.
- **Phase 3 — Advanced:** distributed systems depth, real architecture patterns, trade-off heavy topics.
- **Phase 4 — Interview-Ready:** the standard HLD question set, mock-interview style, plus a repeatable framework and India / UAE / Saudi panel expectations.

Each chapter, once generated, follows the same 8-part structure so you always know what to expect: concept from first principles → real-world case study → diagrams → trade-offs table → common interview questions → practice exercises → curated YouTube videos (English + Hindi) → key takeaways.

Study tip

Do not skip phases. Interviewers at Tier-1 companies (Flipkart, Swiggy, Razorpay, PhonePe, Careem, Noon, STC-tier teams) expect fundamentals to be rock solid before they care about your answer on a 'design WhatsApp' question. Weak fundamentals are the #1 reason experienced engineers fail HLD rounds.

File organization

```
HLD-Guide/  
  Learning_Guide_HLD.pdf          <- this file (master index)  
  Phase-1-Fundamentals/  
    Phase-1-Index.pdf  
    Chapter-01-.pdf  
    Chapter-02-.pdf  
    ...  
  Phase-2-Intermediate/  
    Phase-2-Index.pdf  
    Chapter-01-.pdf ...  
  Phase-3-Advanced/  
    Phase-3-Index.pdf  
    Chapter-01-.pdf ...  
  Phase-4-Interview-Ready/  
    Phase-4-Index.pdf  
    Chapter-01-.pdf ...
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Program Summary

Phase	Focus	Chapters	Est. Hours
Phase 1	Fundamentals	15	~19
Phase 2	Intermediate	15	~22
Phase 3	Advanced	16	~24
Phase 4	Interview-Ready	18	~31
Total		64	~96

Estimates assume focused study, including re-reading and attempting practice exercises. Video links are optional deepening, not required to progress.

Phase 1 — Fundamentals

Goal: build every foundational block from zero. No prior HLD assumed.

#	Chapter	Time
1	What Is System Design (HLD vs LLD) HLD vs LLD, why interviews test it, how it differs from coding rounds	1 hr
2	Client-Server Model & Request Lifecycle How a request travels from browser/app to server and back	1.5 hr
3	DNS & How the Internet Really Works Domain resolution, IP routing, TTLs	1 hr
4	Networking Essentials TCP/IP vs UDP, HTTP/HTTPS, WebSockets	2 hr
5	APIs & Communication Contracts REST, gRPC basics, sync vs async communication	1.5 hr
6	Databases 101 — Relational RDBMS fundamentals, indexes, transactions, ACID	2 hr
7	Databases 101 — NoSQL Key-value, document, column-family, graph stores	2 hr
8	Caching Fundamentals Cache-aside, write-through, write-back, Redis/Memcached	1.5 hr
9	Load Balancing Basics L4 vs L7, algorithms, health checks	1.5 hr
10	Proxies & CDNs Forward vs reverse proxy, CDN mechanics	1 hr
11	Object / Blob Storage Basics S3-style storage, when files don't belong in a database	1 hr
12	Introduction to Scalability Vertical vs horizontal scaling, stateless vs stateful	1.5 hr
13	Non-Functional Requirements & Latency Numbers SLA / SLO / SLI, latency numbers every engineer should know	1.5 hr
14	Back-of-the-Envelope Estimation Estimating QPS, storage, bandwidth for interview problems	1.5 hr
15	Phase 1 Recap & Mini System Designing a simple blog / pastebin end to end	1.5 hr

Phase total: ~19 hours

Phase 2 — Intermediate

Goal: scalability, availability, consistency, messaging, sharding, microservices basics.

#	Chapter	Time
1	Scalability Patterns Deep Dive Read replicas, connection pooling, stateless scaling in practice	1.5 hr
2	Availability & Reliability Uptime SLAs, redundancy, failover, single points of failure	1.5 hr
3	Consistency Models Strong vs eventual consistency, read-your-writes, causal consistency	2 hr
4	CAP Theorem & PACELC Trade-offs during partitions, why 'pick 2 of 3' is oversimplified	1.5 hr
5	Database Replication Leader-follower, multi-leader, quorum-based replication, lag	2 hr
6	Database Sharding & Partitioning Horizontal partitioning, shard keys, hotspots, resharding	2 hr
7	Consistent Hashing Why it matters for sharding/caching, virtual nodes, rebalancing	1.5 hr
8	Message Queues & Async Processing Kafka/RabbitMQ/SQS basics, pub-sub vs point-to-point	2 hr
9	Idempotency & Exactly-Once Delivery Why retries break systems, idempotency keys	1.5 hr
10	Rate Limiting & Throttling Token bucket, leaky bucket, sliding window algorithms	1.5 hr
11	Introduction to Microservices Monolith vs microservices trade-offs, service boundaries	2 hr
12	API Gateways & Service Discovery Gateway responsibilities, registry patterns	1.5 hr
13	Search & Indexing Basics Inverted index, groundwork for autocomplete/search	1.5 hr
14	Data Modeling for Scale Denormalization, polyglot persistence	1.5 hr
15	Phase 2 Recap & Mini System Scalable URL shortener with sharding + caching + rate limiting	2 hr

Phase total: ~22 hours

Phase 3 — Advanced

Goal: distributed systems depth, real architecture patterns, trade-off heavy topics.

#	Chapter	Time
1	Distributed Systems Fundamentals Fallacies of distributed computing, partitions, clock skew	2 hr
2	Consensus Algorithms Paxos/Raft at a conceptual level, leader election	2 hr
3	Distributed Coordination Services Zookeeper/etcd, why they exist	1.5 hr
4	Distributed Transactions 2PC, Sagas, eventual consistency in multi-service writes	2 hr
5	Advanced Data Stores Cassandra/HBase, time-series DBs, Elasticsearch	2 hr
6	OLAP vs OLTP & Data Warehousing Analytical vs transactional workloads, ETL basics	1.5 hr
7	Batch vs Stream Processing MapReduce concepts, Kafka Streams/Flink	2 hr
8	Event-Driven Architecture Event sourcing, CQRS, choreography vs orchestration	2 hr
9	Real-Time Systems & Streaming WebSockets at scale, long polling, SSE	2 hr
10	Bloom Filters & Probabilistic Data Structures Bloom filters, HyperLogLog, count-min sketch	1.5 hr
11	Geo-Distribution & Multi-Region Design Active-active vs active-passive, latency-aware routing	2 hr
12	Security at Scale AuthN/AuthZ (OAuth, JWT), encryption, common attack vectors	2 hr
13	Observability — Logging, Metrics, Tracing Distributed tracing, monitoring/alerting design	1.5 hr
14	Fault Tolerance & Resilience Patterns Circuit breakers, bulkheads, retries with backoff	1.5 hr
15	Backup, Disaster Recovery & Deployment Strategies Backups, DR, canary/blue-green deployments	1.5 hr
16	Phase 3 Recap & Mini System Multi-region, fault-tolerant payment ledger system	2 hr

Phase total: ~24 hours

Phase 4 — Interview-Ready

Goal: the standard HLD interview question set, mock-interview style, with a repeatable framework and India / UAE / Saudi panel expectations.

#	Chapter	Time
1	The HLD Interview Framework Repeatable approach: requirements → estimation → API → HLD → deep dive → trade-offs	1.5 hr
2	India vs UAE vs Saudi Arabia Panel Expectations Flipkart/Swiggy/Razorpay-tier vs Careem/Noon-tier vs STC/Tamara/Jahez-tier panels	1.5 hr
3	Design a URL Shortener Classic warm-up problem, full mock-interview walkthrough	1.5 hr
4	Design a Rate Limiter Applying Phase 2 concepts in an interview setting	1.5 hr
5	Design a Distributed Cache Cache eviction, consistency, scaling cache layers	1.5 hr
6	Design a Typeahead / Autocomplete System Trie-based search suggestions	1.5 hr
7	Design WhatsApp / Chat System Real-time messaging, delivery guarantees, presence, group chats	2 hr
8	Design a News Feed Fan-out on write vs read, ranking, feed generation at scale	2 hr
9	Design Twitter/X Timeline generation, trends, celebrity fan-out problem	2 hr
10	Design an E-Commerce Checkout Inventory consistency, order flow, payment integration touchpoints	2 hr
11	Design a Ride-Hailing / Dispatch System Geo-indexing, matching algorithms, real-time location tracking	2 hr
12	Design a Payment Gateway Idempotency, reconciliation, ledger design, fraud checks	2 hr
13	Design a Food Delivery System Multi-sided marketplace design, ETA prediction, order state machine	2 hr
14	Design a Ticket Booking System High-concurrency seat locking (BookMyShow/IRCTC-style)	2 hr
15	Design a Notification System Fan-out, delivery channels, retry/dedup logic	1.5 hr
16	Design a Distributed File Storage Chunking, metadata service, sync conflict resolution	1.5 hr
17	Design a Web Crawler Crawling at scale, dedup, politeness policies	1.5 hr

#	Chapter	Time
18	Mock Interview Drill Set + Common Pitfalls Timed practice prompts, red flags, handling 'I don't know'	2 hr

Phase total: ~31 hours

Program Workflow

- 1 Study chapters in order within a phase — later chapters assume earlier ones.
- 2 Read the concept section slowly on the first pass; re-read once after attempting the practice exercises.
- 3 Attempt every practice exercise before checking the model answer outline — the struggle is where learning happens.
- 4 Watch at least one English and one Hindi video per chapter if the written explanation didn't fully click.
- 5 After finishing a phase, redo its Recap & Mini System chapter from memory, without notes.
- 6 Move to the next phase only when you can explain every chapter's key takeaways out loud, unaided.

Where you are now

Phase 1 is approved and generated. Phases 2–4 will be generated only when you explicitly approve each one.